(a) Using mathematical induction, prove that, for all positive integers n,

$$\cos x \cos(2x)\cos(4x)\ldots\cos(2^n x)=\frac{\sin(2^{n+1}x)}{2^{n+1}\sin x},$$

where x is not a multiple of π.

(b) Using (a), evaluate $\sum_{k=3}^{20}\log_2\left|\cos\left(2^k\cdot\frac{\pi}{3}\right)\right|$.

(7 marks)

a) let $P(n): \cos x\cdot\cos(2x)\cdot\cos(4x)\cdots\cos(2^n x)=\frac{\sin(2^{n+1})x}{2^{n+1}\sin x}$

for $n\in\mathbb{Z}^+$ & x is not a multiple of π

when $n=1$, $LHS=\cos x\cos 2x$

$RHS=\frac{\sin 4x}{4\sin x}=\frac{\sin 2x\cos 2x}{2\sin x}=\frac{\sin x\cos x\cos 2x}{\sin x}=\cos x\cos 2x$

$LHS=RHS$, $\therefore P(1)$ is true. 1

assume $P(k)$ is true for some $k\in\mathbb{Z}^+$.

i.e. $\cos x\cdot\cos 2x\cdot\cos 4x\cdots\cos 2^k x=\frac{\sin(2^{k+1})x}{2^{k+1}\sin x}$ 1M

when $n=k+1$,

$LHS=\cos x\cdot\cos 2x\cdot\cos 4x\cdot\cos 2^k x\cdot\cos 2^{k+1}x$

$=\frac{\sin(2^{k+1})x}{2^{k+1}\sin x}\cdot\cos 2^{k+1}x$ 1M

$=\left(\frac{1}{2^{k+1}\sin x}\right)\left(\frac{1}{2}\sin(2(2^{k+1})x)\right)$ $\sin 2a=2\sin a\cos a$

$=\frac{\sin(2^{k+2})x}{2^{k+2}\sin x}=\frac{\sin(2^{(k+1)+1})x}{2^{(k+1)+1}\sin x}=RHS$.

therefore, $P(k+1)$ is also true. 1

by the principle of mathematical induction, $P(n)$ is true for $n\in\mathbb{Z}^+$ and x is not a multiple of π.

Answers written in the margins will not be marked.

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b) required value

$$= \log_2\left[\cos\left(2^3 \cdot \frac{\pi}{3}\right)\right] + \log_2\left[\cos\left(2^4 \cdot \frac{\pi}{3}\right)\right] + \dots + \log_2\left[\cos\left(2^{20} \cdot \frac{\pi}{3}\right)\right]$$

$$= \log_2\left(\prod_{k=3}^{20} \cos\left(2^k \cdot \frac{\pi}{3}\right)\right)$$ 1M log properties

$$= \log_2\left(\frac{\dfrac{\sin\left(2^{20+1}\left(\frac{\pi}{3}\right)\right)}{2^{20+1}\sin\left(\frac{\pi}{3}\right)}}{\dfrac{\sin\left(2^{2+1}\left(\frac{\pi}{3}\right)\right)}{2^{2+1}\sin\left(\frac{\pi}{3}\right)}}\right)$$ 1M

$$= \log_2\left(\frac{2097152^{-1}}{8^{-1}}\right)$$

$= -18$ // 1A

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